SINGAPORE DIMENSIONS • GRADE 1A • CHAPTER 7

Numbers to 20: Place Value & Staging

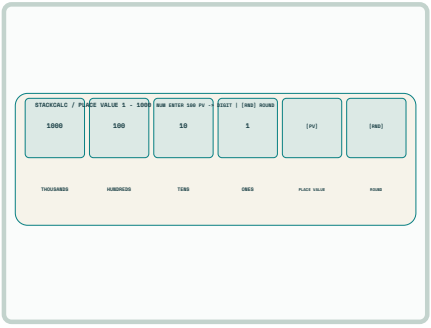
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.NBT.B.2

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Place Value Tray Whole

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
One 10-rod alongside 4 unit cubes shows 14 as 1 ten and 4 ones.

Sketch Model on Paper:
Draw a 10-rod bar and 4 individual unit squares.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	1	Step 1	Execute [1]
2.	0	Step 2	Execute [0]
3.	ENTER	Step 3	Execute [ENTER]
4.	6	Step 4	Execute [6]

EXPECTED LCD X: 16 (1 ten, 6 ones)

1 **Combine 1 ten (10) and 3 ones (3).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Combine 1 ten (10) and 7 ones (7).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Decompose 18: subtract 8 ones. What remains?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Decompose 15: subtract 1 ten (10). What remains?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 7

Numbers to 20: Place Value & Staging • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Place Value Studio (Base-10 Rods & Units)

CAD Blueprint: place_value_tray_whole

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/place_value_tray_whole.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal Tray & White Numerals (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Combine 1 ten (10) and 3 ones (3).** Standard: CCSS.MATH.1.NBT.B.2	13	[1][0] [ENTER] [3] [+]
2	**Combine 1 ten (10) and 7 ones (7).** Standard: CCSS.MATH.1.NBT.B.2	17	[1][0] [ENTER] [7] [+]
3	**Decompose 18: subtract 8 ones. What remains?** Standard: CCSS.MATH.1.NBT.B.2	10 (1 ten)	[1][8] [ENTER] [8] [-]
4	**Decompose 15: subtract 1 ten (10). What rema** Standard: CCSS.MATH.1.NBT.B.2	5 (5 ones)	[1][5] [ENTER] [1][0] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for reversing digits (writing 41 instead of 14).

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.